

N.S. LABORATORY SDN. BHD.

ABZOLE, 100 mg/ml, Liquid suspension

Active Ingredient:

Each ml contains
Albendazole-----100mg

Preservative:

Each ml contains
Methyl paraben-----1mg

Product Description:

White or slightly yellowish colour liquid suspension

Dosage Form:

Liquid suspension

Pharmacodynamics:

Benzimidazoles bind to nematode tubulin, a protein necessary for the formation and viability of microtubules. This occurs primarily in absorptive intestinal cells resulting in the absence of microtubules in the intestinal cells of the nematode, with the result that these cells cannot absorb nutrients, thus causing a consequent reduction in glycogen and effective starvation of the parasites. Structural differences have been shown to exist between tubulin from mammalian and helminth sources, resulting in the preferential toxicity of albendazole to the helminth and not to the host.

Benzimidazoles have also been shown to inhibit the fumarate reductase system of helminths and impair energy production.

Pharmacokinetics:

Albendazole has poor water solubility and limited absorption from the gastrointestinal tract (about 50% of the oral dose is absorbed in cattle). Following absorption, there is rapid first pass metabolism in the liver and the sulphide moiety of albendazole is oxidised to the pharmacologically active sulphoxide, then to the sulphone, followed by deacetylation of the carbamate group to form the 2-aminosulphone.

Indication:

A broad spectrum multi-purpose anthelmintic for the control of mature and developing immature forms of gastrointestinal roundworms, lungworms, tapeworms and adult liver fluke in cattle and sheep. The product is also ovicidal against fluke and roundworm eggs, thus reducing pasture contamination.

In cattle the product is active against the following species:

Roundworms: *Ostertagia*, *Haemonchus*, *Trichostrongylus*, *Nematodirus*, *Oesophagostomum*, *Bunostomum*, *Cooperia* and *Strongyloides* spp. It is usually effective against inhibited larvae of *Cooperia* and *Ostertagia*,

Lungworms: *Dictyocaulus viviparus*,

Tapeworms: *Moniezia* spp.,

Adult liver fluke: *Fasciola hepatica*

In sheep, the product is active against benzimidazole-susceptible strains of the following species:

Roundworms: *Ostertagia*, *Haemonchus*, *Trichostrongylus*, *Nematodirus*, (including *N.battus*), *Chabertia* and *Oesophagostomum*.

It is usually effective against inhibited larvae of *Ostertagia*.

Lungworms: *Dictyocaulus filaria*,

Tapeworms: *Moniezia* spp.,

Adult liver fluke: *Fasciola hepatica*

Mode of Administration:

Oral administration.

Recommended Dosage:

For oral administration only using properly calibrated dosing equipment. To ensure administration of a correct dose, bodyweight should be determined as accurately as possible; accuracy of the dosing device should be checked. If animals are to be treated collectively rather than individually, they should be grouped according to their bodyweight and dosed accordingly, in order to avoid under- or over-dosing. One ml contains 100mg albendazole. Shake the container before use.

Cattle- Worm dose: For the control of roundworms, lungworms, tapeworms and fluke and roundworm eggs.

Dosage: 7.5 mg albendazole per kg bodyweight.

(7.5 ml product per 100 kg bodyweight)

Fluke and Worm Dose: For the additional treatment of adult liver fluke (chronic fascioliasis) in cattle.

Dosage: 10 mg albendazole per kg bodyweight.

(10 ml product per 100 kg bodyweight)

Sheep- Worm dose: For the control of roundworms, lungworms, tapeworms and fluke and roundworm eggs.

Dosage: 5 mg albendazole per kg bodyweight.

(1 ml product per 20 kg bodyweight)

Fluke and Worm Dose: For the additional treatment of adult liver fluke (chronic fascioliasis) in sheep.

Dosage: 7.5 mg albendazole per kg bodyweight.

(1.5 ml product per 20 kg bodyweight)

Contraindications:

Do not use in case of hypersensitivity to the active substance or to any of the excipients.

Special Warnings for each Target Species:

Cattle suffering from severe lung damage due to heavy lungworm infestation may continue to cough for some weeks after infection. Care should be taken to avoid the following practices because they increase the risk of development of resistance and could ultimately result in ineffective therapy:

- Too frequent and repeated use of anthelmintics from the same class, over an extended period of time.

- Underdosing, which may be due to underestimation of bodyweight, misadministration of the product, or lack of calibration of the dosing device (if any).

Suspected clinical cases of resistance to anthelmintics should be further investigated using appropriate tests (e.g. Faecal Egg Count Reduction Test). Where the results of the test(s) strongly suggest resistance to a particular anthelmintic belonging to another pharmacological class and having a different mode of action should be used. Resistance to benzimidazoles (which includes albendazole) has been reported in *Teladorsagia*, *Haemonchus*, *Cooperia* and *Trichostrongylus* species in small ruminants in a number of countries. Resistance to albendazole has been reported in *Cooperia* and *Teladorsagia* species in cattle in developed countries such as New Zealand. Therefore the use of this product should be based on local (regional, farm) epidemiological information about susceptibility of nematodes and recommendations on how to limit further selection for resistance to anthelmintics.

Special Precautions for Use in Animals:

Care must be taken not to damage the pharyngeal region when dosing, particularly in sheep. Not to be diluted or mixed with other products. Avoid the introduction of contamination during use. Intensive use or misuse of anthelmintics can give rise to resistance. To reduce this risk dosing programmes should be discussed with veterinary surgeon. If in doubt, consult a veterinary surgeon.

Special Precautions to be taken by the Person Administering the Medicinal Products to Animals:

This product can cause skin and eye irritation. Personal protective equipment consisting of suitable protective clothing, including impermeable rubber gloves, should be worn when handling the product. In case of accidental spillage onto skin, wash the affected area with soap and water. If irritation persists seek medical advice immediately and show the package leaflet or the label to the physician. In case of accidental eye exposure, flush eye thoroughly with running water. If irritation persists seek medical advice immediately and show the package leaflet or the label to the physician. Wash hands after use.

Interactions with other Medicaments:

None known.

Usage during Pregnancy and Lactation:

Pregnancy- Do not dose ewes at the "flake and worm" dose rate, (7.5 mg/kg) during tupping or for 1 month after removing the rams.

Lactation- Can be safely used during lactation.

Fertility- The use in breeding bulls or pregnant cattle is not expected to interfere with their reproductive performance.

Adverse Effects:

None

Overdose and Treatment:

Do not exceed the stated dose.

Withdrawal Period (Meat and Offal):

Cattle- 14 days; Milk- 60 hours

Sheep- 4 days

Do not use in sheep producing milk for human consumption.

Storage Condition:

Store in cool, dark place (< 30°C). Protect from light.

Shelf Life:

36 months

Shelf Life after Opening:

Discard after 24 hours

Packaging Available:

1L/bottle; 12 bottles/carton; 12kg/carton

Manufacturer and Registration Holder:

N.S. Laboratory Sdn. Bhd.

54, Jalan S2C2, Green Technology Park, Seremban 2, 70300 Seremban, Negeri Sembilan, Malaysia.

Revised Date:

26/06/2024